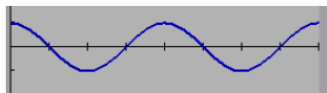
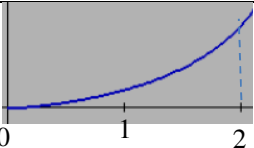
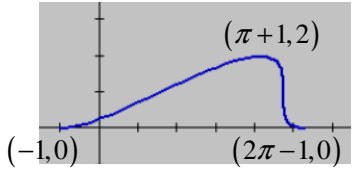
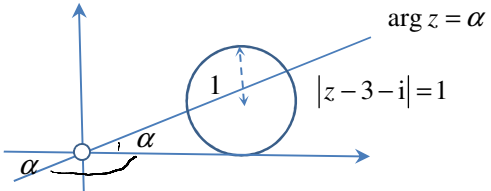


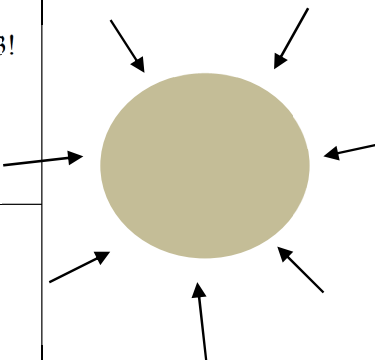
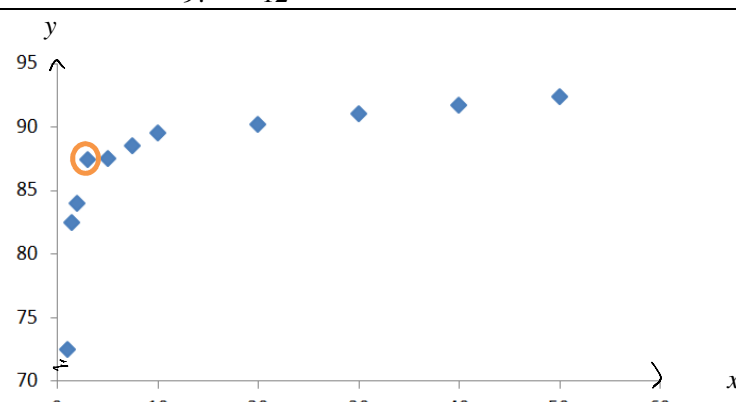
Qn	Solution	Comments
1	Let the depth of water at t minutes be h m. $\tan \alpha = 0.5 = \frac{r}{h} \Rightarrow r = \frac{h}{2}$ $V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}$ $\frac{dV}{dh} = \frac{\pi h^2}{4}$ When $V = 3\text{m}^3$, $h = \sqrt[3]{\frac{36}{\pi}}$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{4}{\pi h^2} \times \frac{1}{10} = \frac{2}{5\pi h^2}$ When $h = \sqrt[3]{\frac{36}{\pi}}$, $\frac{dh}{dt} = \frac{2}{5\pi \left(\frac{36}{\pi}\right)^{\frac{2}{3}}} = 0.0251 \text{ m min}^{-1}$	It is easier to write V in terms of h rather than r. Ask students to give a reason. $\frac{dh}{dt}$ should be a value without the unit. Must always check if the letters are constants or variables Cambridge accept answers in the range [0.025, 0.0251]
2ai	$\int x^2 \cos nx \, dx = x^2 \frac{\sin nx}{n} - \int \frac{2x \sin nx}{n} \, dx$ $= x^2 \frac{\sin nx}{n} - \left(\frac{2x}{n} \left(-\frac{\cos nx}{n} \right) - \int \frac{2}{n} \left(-\frac{\cos nx}{n} \right) dx \right)$ $= \frac{x^2 \sin nx}{n} + \frac{2x \cos nx}{n^2} - \int \frac{2 \cos nx}{n^2} \, dx$ $= \frac{x^2 \sin nx}{n} + \frac{2x \cos nx}{n^2} - \frac{2 \sin nx}{n^3} + C$	Be careful with the sign when integrating for this part. Use brackets for the second application of 'by parts'.
ii	$\int_{\pi}^{2\pi} x^2 \cos nx \, dx = \left[\frac{x^2 \sin nx}{n} + \frac{2x \cos nx}{n^2} - \frac{2 \sin nx}{n^3} \right]_{\pi}^{2\pi}$ $= \left(0 + \frac{2(2\pi)}{n^2} - 0 \right) - \left(0 + \frac{2\pi \cos n\pi}{n^2} - 0 \right)$ $= \frac{4\pi}{n^2} - \frac{2\pi \cos n\pi}{n^2}$ When n is an odd integer, $\int_{\pi}^{2\pi} x^2 \cos nx \, dx = \frac{4\pi}{n^2} - \frac{2\pi \cos n\pi}{n^2} = \frac{4\pi}{n^2} + \frac{2\pi}{n^2} = \frac{6\pi}{n^2}$ When n is an even integer, $\int_{\pi}^{2\pi} x^2 \cos nx \, dx = \frac{4\pi}{n^2} - \frac{2\pi \cos n\pi}{n^2} = \frac{4\pi}{n^2} - \frac{2\pi}{n^2} = \frac{2\pi}{n^2}$ $\therefore$ The possible values of a are 2 and 6.	Note the hint in the question: '...giving your answers in the form $a \frac{\pi}{n^2}$, where the possible values of a are to be determined.' There are 2 answers. From $\frac{4\pi}{n^2} - \frac{2\pi \cos n\pi}{n^2}$, to get to the form of $a \frac{\pi}{n^2}$, need to evaluate $\cos n\pi$ and note that n is a positive integer. Look at the graph of $y = \cos x$  When x is an even multiple of π, $\cos x = 1$, and when x is an odd multiple of π, $\cos x = -1$.

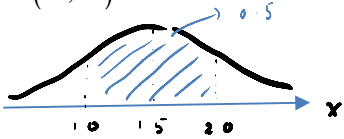
b		
	Volume = $\pi \int_0^2 y^2 dx$ $= \pi \int_0^2 \frac{x^3}{(9-x^2)^2} dx$ $= -\frac{\pi}{2} \int_0^2 \frac{x^2(-2x)}{(9-x^2)^2} dx$ $= -\frac{\pi}{2} \int_9^5 \frac{9-u}{u^2} du$ $= -\frac{\pi}{2} \int_9^5 9u^{-2} - \frac{1}{u} du$ $= -\frac{\pi}{2} \left[-9u^{-1} - \ln u \right]_9^5$ $= \frac{\pi}{2} \left(\frac{9}{5} + \ln 5 - 1 - \ln 9 \right)$ $= \pi \left(\frac{2}{5} + \frac{1}{2} \ln \frac{5}{9} \right)$	While the question asks for exact answer, you can still use the GC to check your answer. But using a GC to get the answer in decimals will not meet the requirements of the question.
3i	When $y = 0$, $1 - \cos t = 0 \Rightarrow t = 0$ or 2π $t = 0 \Rightarrow x = -1$ $t = 2\pi \Rightarrow x = 2\pi - 1$ $\frac{dx}{dt} = 1 + \sin t$, $\frac{dy}{dt} = \sin t$ $\frac{dy}{dx} = \frac{\sin t}{1 + \sin t}$ When $\frac{dy}{dx} = 0$, $\sin t = 0 \Rightarrow t = 0, \pi, 2\pi$ When $t = \pi$, $x = \pi + 1$, $y = 2$ 	Read the question. We need the exact coordinates of the maximum point and the points of intersection with the x-axis. We need to show working. We do not need to find the coordinates of point of intersection with the y-axis. Notice how the shape should look like at the 2 end-points and how the graph between the maximum point and the right end-point should not curl to the left to yield 2 images at one x value.

ii	$\begin{aligned} \text{Area} &= \int_{-1}^{a-\cos a} y \, dx \\ &= \int_0^a (1 - \cos t)(1 + \sin t) \, dt \\ &= \int_0^a 1 - \cos t + \sin t - \sin t \cos t \, dt \\ &= \int_0^a 1 - \cos t + \sin t - \frac{\sin 2t}{2} \, dt \\ &= \left[t - \sin t - \cos t + \frac{\cos 2t}{4} \right]_0^a \\ &= a - \sin a - \cos a + \frac{\cos 2a}{4} + 1 - \frac{1}{4} \\ &= a - \sin a - \cos a + \frac{\cos 2a}{4} + \frac{3}{4} \end{aligned}$	
iii	When $t = \frac{\pi}{2}$, $\frac{dy}{dx} = \frac{1}{2}$ Gradient of normal = -2 $x = \frac{\pi}{2}, y = 1$ $y - 1 = -2\left(x - \frac{\pi}{2}\right)$ $y = -2x + \pi + 1$ Equation of normal is $y = -2x + \pi + 1$ When $x = 0$, $y = 1 + \pi$ When $y = 0$, $x = \frac{1 + \pi}{2}$ $\text{Area of triangle } OEF = \frac{1}{2}(\pi + 1)\left(\frac{\pi + 1}{2}\right) = \frac{(\pi + 1)^2}{4} \text{ units}^2$	
4ai	Im z_2 z_1 $(3, 1)$ O Re  $z - 3 - i = 1$ $z - (3 + i) = 1$	Not in H2 Math syllabus $\arg z = \alpha$ is actually made up of 2 half lines. There are 2 values of α such that $\tan \alpha = 0.4$. In exam report there was no mention of α when it is in 3rd quad. Draw the diagram with equal scale on both axes. The relative positions of the half line and circle must be correct. Check: Does your line pass through the centre of the circle? If not, should it be above or below the centre of the circle?

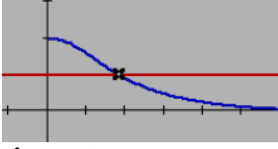
		Is the real axis tangential to your circle?
ii	Equation of half line is $y = 0.4x$ Equation of circle is $(x-3)^2 + (y-1)^2 = 1$ By GC, $z_1 = 2.0187 + 0.80748i$ and $z_2 = 3.8434 + 1.5373i$ Mid point of segment joining z_1 and z_2 is $(2.9311, 1.1724)$. $y - 1.1724 = \frac{-1}{0.4}(x - 2.9311)$ $y = -2.5x + 8.5002$ Cartesian equation of $z - z_1 = z - z_2$ is $y = -2.5x + 8.5002$ By GC, the 2 values of z are $2.63 + 1.93i$ and $3.37 + 0.716i$.	Not in H2 Math syllabus Question did not mention that we need to find the exact coordinates. We can find the coordinates of points of intersection via intersection of graphs in the GC.
bi	Let $z^3 = w$ $z^3 = \sqrt{8}e^{-\frac{\pi}{4}i} = \sqrt{8}e^{\left(-\frac{\pi}{4} + 2k\pi\right)i}, k \in \mathbb{Z}$ $z = \sqrt[3]{2}e^{\left(\frac{(8k-1)\pi}{12}\right)i}, k = 0, \pm 1$ $= \sqrt{2}e^{-\frac{3\pi}{4}i}, \sqrt{2}e^{-\frac{\pi}{12}i}, \sqrt{2}e^{\frac{7\pi}{12}i}$ The cube roots of w are $\sqrt{2}e^{-\frac{3\pi}{4}i}, \sqrt{2}e^{-\frac{\pi}{12}i}$ and $\sqrt{2}e^{\frac{7\pi}{12}i}$.	Not in H2 Math syllabus
ii	$\arg(w^* w^n) = \frac{\pi}{4} + n\left(-\frac{\pi}{4}\right) + 2k\pi = \frac{\pi}{4}(1-n) + 2k\pi, k \in \mathbb{Z}$ $\frac{\pi}{4}(1-n) + 2k\pi = \frac{\pi}{2}$ When n is a positive whole number, $\frac{\pi}{4}(1-n) = \frac{\pi}{2} - 2k\pi = -\frac{3}{2}\pi, -\frac{7}{2}\pi, -\frac{11}{2}\pi, \dots$ $1-n = -6, -14, -22, \dots$ $n = 7, 15, 23, \dots$ Smallest positive whole number value of n is 7.	Not in H2 Math syllabus
5i	$P(\text{player wins}) = \frac{1}{7} \times \frac{3}{6} + \frac{2}{7} \times \frac{2}{6} + \frac{4}{7} \times \frac{1}{6}$ $= \frac{11}{42}$	
ii	$P(\text{Blplayer wins}) = \frac{\frac{2}{7} \times \frac{2}{6}}{\frac{11}{42}}$ $= \frac{4}{11}$	
iii	Required prob = $\left(\frac{1}{7} \times \frac{3}{6}\right) \times \left(\frac{2}{7} \times \frac{2}{6}\right) \times \left(\frac{4}{7} \times \frac{1}{6}\right) \times 3!$ $= \frac{4}{1029}$	The game could be won in a total of $3!$ ways. So we need to multiply by $3!$ Why is the answer not $\left(\frac{11}{42}\right)^3$, i.e cube of the ans in (i)?

6ia	Number of males = $2345 + 1013 + 237 + 344 = 3939$ Total number of employees = 6599 Number of males in the sample = $\frac{3939}{6599} \times 100 = 59.7 \approx 60$	Not in H2 Math syllabus Is there a need to work out the number of males from each department?
b	Number of females from development department in the sample $= \frac{679}{6599} \times 100 = 10.2 \approx 10$	Not in H2 Math syllabus
ii	The stratified sample from (i) should not be used since an employee is not chosen independently of all other employees in such a sample given that a pre-determined number is to be chosen from each stratum. Need to modify this question for new syllabus and write out a solution that is easier for student to understand. Have modified this question for new syllabus – refer to 2017 Mock Exam Paper 2	Not in H2 Math syllabus The most common wrong answer is that the strata should be based on age rather than on gender and department.
iii	Let X years be the age of an employee and μ years be the population mean age of all the employees $H_0: \mu = 37$ $H_1: \mu < 37$ Level of significance: 5% Test statistics: Since $n = 80$ is large, by Central Limit Theorem, $\bar{X}$ is approximately normal. When H_0 is true, $Z = \frac{\bar{X} - 37}{\sqrt{\frac{140}{n}}} \sim N(0,1)$ approximately. For H_0 to be rejected, $P(Z < z) \leq 0.05$ $\frac{\bar{x} - 37}{\sqrt{\frac{140}{80}}} \leq -1.6449$ $\bar{x} \leq 34.824$ Set of possible mean age of the sample is $\{\bar{x} : \bar{x} \leq 34.8\}$.	Note that the sample size is different from (i) Some weaker student could use $n - 1$ instead of n in the denominator of the test statistic. They may also be confused over standard deviation vs variance.
iv	When $\bar{x} = 35.2$, $p\text{-value} = 0.086809$ For H_0 to not be rejected, $0.086809 > \frac{\alpha}{100}$ Set of possible values of α is $\{\alpha \in \mathbb{R} : \alpha < 8.68\}$.	Reminder: If a wrong answer is provided using a GC without showing any working, no credit can be awarded. Cambridge accepts 8.69.
7i	Number of ways = ${}^4P_3 = 24$	Why is the answer not 4C_3 ?
ii	Number of ways = ${}^{10}P_3 - {}^6P_3 - 24 = 576$ Alternatively Number of ways = ${}^3C_2 \times {}^4P_2 \times {}^6P_1 + {}^3C_2 \times {}^4P_1 \times {}^6P_2 = 576$ Alternatively, Number of ways = $({}^6C_2 \times {}^4C_1 + {}^4C_2 \times {}^6C_1) \times 3! = 576$	The first method uses complement approach. The second approach is by cases. However, we have to multiply by 3C_2 because we need to choose the 2

		positions that are filled by the same gender. The last approach is to pick the group first then consider the various positions.
iii	No. of ways to arrange 10 people round a table = $9!$ Treat the chairperson, secretary & treasurer as 1 unit, and arrange the 8 units round a table = $7!$ For each of the $7!$ arrangements, the 3 can arrange themselves in $3!$ ways. Thus number of arrangements = $7! \times 3!$ Req prob = $\frac{7!3!}{9!} = \frac{1}{12}$	Parts iii & iv are probability questions 
iv	Arrange the remaining 7 people round a table = $6!$ No. of ways to slot in the chairperson, secretary & treasurer = ${}^7C_3 \times 3! = {}^7P_3$ Req. prob = $\frac{6! \times {}^7P_3}{9!} = \frac{5}{12}$	
8i		Need to circle the point that corresponds to the wrong entry. Need to use the graph paper. Need to use the suggested scales.
ii	From the scatter diagram, y increases by decreasing amounts as x increases, the relationship between x and y should not be related by a linear equation of the form $y = ax + b$. Alternatively, The points on the scatter diagram do not lie close to a straight line. Hence the relationship between x and y should not be related by an equation of the form $y = ax + b$.	
iii	As x increases, $\frac{1}{x}$ decreases. For $\frac{c}{x}$ to be increasing as x increases, c is negative. Alternatively, $\frac{dy}{dx} = \frac{-c}{x^2}$ and the scatter plot shows y to be increasing as x increases. Hence c is negative so that the gradient is positive. For the graph of $y = \frac{c}{x} + d$ to have a horizontal asymptote above the x-axis, d is positive.	Another wrong approach would be to try to discuss where the curve cuts the y-axis. 'c must be negative and d positive to give the shape of the scatter diagram' is not accepted.

iv	From the GC, $r = -0.979546 = -0.980$ (to 3 s.f.) $c = -17.484 = -17.5$ (to 3 s.f.) $d = 91.750 = 91.8$ (to 3 s.f.)	Error arose due to the tendency to forget to exclude the incorrect entry or in identifying the incorrect entry wrongly.
v	$y = \frac{-17.484}{x} + 91.750$ When $x = 3$, $y = 85.922 = 85.9$ (to 3 s.f.) The estimate is reliable since (1) $r = -0.980$ is close to -1 and from the scatter diagram indicates that the model $y = \frac{c}{x} + d$ where $c < 0$ and $d > 0$ is appropriate, & (2) $x = 3$ is inside the data range of $1 \leq x \leq 50$.	Cambridge accepts 86.0.
9a	$X \sim N(15, a^2)$  Given $P(10 < X < 20) = 0.5$ $P\left(\frac{10-15}{a} < Z < \frac{20-15}{a}\right) = 0.5$ From the GC, $P(-0.67449 < Z < 0.67449) = 0.5$ $-\frac{5}{a} = -0.67449$ $a = 7.4130 \approx 7.41$	Note that the mid point of 10 and 20 is the mean 15. Hence we can make use of <code>invNorm(0.5, 0, 1, Centre)</code> Use a diagram as a thinking tool, especially for the more visual learners. Significant number of students omitted this part.
b	$Y \sim B(4, p)$ $P(Y=1) + P(Y=2) = 0.5$ ${}^4C_1 p(1-p)^3 + {}^4C_2 p^2(1-p)^2 = 0.5$ $4p(1-3p+3p^2-p^3) + 6p^2(1-2p+p^2) = 0.5$ $4p-12p^2+12p^3-4p^4+6p^2-12p^3+6p^4 = 0.5$ $4p-6p^2+2p^4 = 0.5$ $4p^4-12p^2+8p = 1$ By GC, $p = -2.01, 1.25, 0.599, 0.166$ Since $0 \leq p \leq 1$, $p = 0.599$ or 0.166.	
c	Let W be the number of questions George guesses correctly, out of 100. $W \sim B\left(100, \frac{1}{3}\right)$ Since n is large, $np = \frac{100}{3} > 5$, $n(1-p) = \frac{200}{3} > 5$	Not in H2 Math syllabus Error arose due to truncated values for the mean or standard deviation.

	$W \sim N\left(\frac{100}{3}, \frac{200}{9}\right)$ approximately $P(W \geq 30) = P(W > 29.5)$ $= 0.792$	
10i	The average number of the particular type of weed per square unit area of Mia's field is a constant. The growth of a weed of the particular type is independent of the growth of all other weeds of that type.	Not in H2 Math syllabus Common wrong answers include: 1. "The weed grows independently of other types of weeds". (Should confine to weeds of that particular type growing independently of one another.) 2. "The average number of weeds in the field is constant". (Should talk about average per square unit of land, that is, the average "rate" and also should confine to weeds of that particular type.)
ii	Let X be the number of dandelion plants in 1m^2 of Mia's field. $X \sim \text{Po}(1.5)$ $P(X \geq 2) = 1 - P(X \leq 1) = 0.442$	Not in H2 Math syllabus
iii	Let Y be the number of dandelion plants in 4m^2 of Mia's field. $Y \sim \text{Po}(6)$ $P(Y \leq 3) = 0.151$	Not in H2 Math syllabus
iv	Let W be the number of dandelion plants in 80m^2 of Mia's field. Method 1: $W \sim \text{Po}(120)$ Since $120 > 10$, $W \sim N(120, 120)$ approximately $P(110 \leq W \leq 140) = P(109.5 \leq W \leq 140.5) = 0.800$ Method 2: $W = X_1 + X_2 + \dots + X_{80}$ By CLT, $W \sim N(120, 120)$ approximately $P(110 \leq W \leq 140) = 0.785$	Not in H2 Math syllabus Note how the word inclusive is used to remove the controversy of the term "between". Note that the answer obtained by CLT without continuity correction is not given by Cambridge. Report did not mention about students using CLT. Wonder if that means continuity correction in such contexts is their standard practice over there.
v	Let A and B be the number of daisies in 1m^2 and 2m^2 of Mia's field respectively. $A \sim \text{Po}(\lambda), B \sim \text{Po}(2\lambda)$ $P(A \leq 2) = P(B > 2)$ $= 1 - P(B \leq 2)$	Not in H2 Math syllabus Not all could simplify $\frac{\lambda^0}{0!}$.

$e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2} \right) = 1 - e^{-2\lambda} \left(1 + 2\lambda + \frac{4\lambda^2}{2} \right)$ $e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2} \right) + e^{-2\lambda} (1 + 2\lambda + 2\lambda^2) = 1$ By GC,  $\lambda = 1.85$	Different functions could be considered on both sides of the equation to solve for λ. Working (graphs in this case) needs to be shown when GC is used.
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